Luis Agustín Cardenas Pena (Curriculum Vitae)

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Papers

I am a mathematician, researcher, and university instructor with over seven years of research experience and ten years of teaching experience at both undergraduate and graduate levels. I earned my Bachelor's degree in Mathematics from the National University of Comahue (Argentina), followed by a Master's degree in Mathematical Sciences from the University of Talca (Chile). In March 2026, I completed my Ph.D. in Mathematics at the University of Talca, where I wrote my dissertation entitled "Schur Bases for the Ring of m-Symmetric Functions."

My research interests lie primarily in algebraic combinatorics, symmetric functions, and tableau theory. Throughout my academic career, I have actively combined research with teaching, viewing these activities as complementary aspects of university life. While research advances mathematical knowledge, teaching and outreach provide opportunities to engage with students and contribute directly to their intellectual development. I believe that excellence in higher education requires a strong commitment to both scholarly research and effective instruction.

Publications

Peer-reviewed articles

[1] Susanna Fishel and **Luis Pena**, "Parking functions with zero secondary *dinv*," *Arabian Journal of Mathematics* **15**(2) (2026), 725–733. doi:10.1007/s40065-025-00562-8.

[2] **Luis Agustín Cárdenas Pena**, "Recortando polígonos, de las tijeras a la generalización" ["Cutting polygons: From scissors to generalizations"], *Revista de Educación Matemática* **35**(2) (2020), 51–69. doi:10.33044/revem.29729. [In Spanish.]

Preprints

[1] Luc Lapointe and **Luis Pena**, "A proof of the m-symmetric Macdonald positivity at $t = 1$," arXiv:2608.10223 [math.CO], 2026.

[2] Luc Lapointe and **Luis Pena**, "A new characterization of right keys, and the m-symmetric Schur functions at $t = 0$," arXiv:2608.13276 [math.CO], 2026.

[3] Luis Pena, "Cauchy identities for skew Ferrers shapes via RSK and keys," arXiv:2609.15852 [math.CO], 2026.

Teaching Experience

As Course Instructor

2026 Instructor of the module Calculus in one variable for the common curriculum of Bioinformatics Engineering and Civil Engineering programs , University of Talca, Chile.

2026 Instructor of the module Algebra for the common curriculum of Civil Mining Engineering, Electrical Engineering, and Computer Engineering programs, University of Talca, Chile.

2026 Instructor of the module Introduction to Mathematics for the common curriculum of Civil Mining Engineering, Electrical Engineering, and Computer Engineering programs, University of Talca, Chile.

2026 Instructor of the module Introduction to Mathematics for the common curriculum of Industrial Civil Engineering, Mechatronics Engineering, and Mechanical Engineering programs, University of Talca, Chile.

2025 Instructor of the module Introduction to Mathematics for the programs in Video Game Development and Virtual Reality Engineering, and Bioinformatics Engineering, University of Talca, Chile.

2025 Instructor of the module Mathematics for Biomedical Sciences for the Medical Technology program, University of Talca, Chile.

2024 Instructor of the module Mathematics for Business for the Business Administration and Management Engineering program, Catholic University of Maule, Chile.

2024 Instructor of the module Mathematics for Medical Technology for the Medical Technology program, Catholic University of Maule, Chile.

2023 Instructor of the Academic Preparation Workshop for Education majors (one-week intensive course), University of Talca, Chile.

As Teaching Assistant

2025	Teaching Assistant, Topics in Topology (Graduate Algebraic Topology), University of Talca, Chile.
2019-2025	Professional Teaching Assistant for Engineering, Biomedical Technology, Nursing, and Accounting programs (17 courses in total), University of Talca, Chile. Courses included Differential Equations, Calculus, Linear Algebra, Precalculus, and Basic Mathematics.
2016	Volunteer Teaching Assistant, National University of Comahue, Argentina. Algebra I for Mathematics majors and teacher-training students.
2016	Volunteer Teaching Assistant, National University of Comahue, Argentina. Introduction to Analysis (Topology of Metric Spaces) for Mathematics majors and teacher-training students.

Conference, School, and Colloquium Presentations

- (December 2025) Schur Bases for m-Symmetric Functions, Annual Meeting of the Argentinian Mathematical Union (UMA), Balseiro Institute, Bariloche, Argentina.
- (December 2025) Parking Functions with Secondary dinv Equal to Zero, Annual Meeting of the Argentine Mathematical Union (UMA), Balseiro Institute, Bariloche, Argentina.
- (August 2025) Schur Bases for m-Symmetric Functions and the RSK Algorithm (poster), elENA X, FAMAF, Córdoba, Argentina.
- (April 2025) Schur Bases for m-Symmetric Functions (oral presentation), XXXVII Southern Zone Mathematics Meeting, Catholic University of Maule, Talca, Chile.
- (August 2024) Macdonald's Conjecture and m-Symmetric Functions (oral presentation), Non-Degenerate Student Seminar, Institute of Mathematics, University of Talca, Chile.
- (March 2024) Schur Bases for m-Symmetric Functions (poster), II ENEMAT 2024, University of Talca, Chile.
- (October 2022) Combinatorial Interpretation of the m-Symmetric Schur Basis (oral presentation), Mini-Symposium on Algebra and Category Theory, Pontifical Xavierian University, Bogotá, Colombia.
- (July 2022) Symmetric Functions and Young Tableaux (short course), First Meeting of Mathematics Students and Graduates of the National University of Comahue, Bariloche, Argentina.

Academic Service

- 2025 – Organizer, *Seminario No-Degenerado*, student seminar, University of Talca, Chile.

2021–2026	Ph.D. in Mathematics, Universidad of Talca, Chile.
	Título de la tesis: ”Schur Bases for the Ring of m-Symmetric Functions.”.
	Director: Luc Lapointe (Algebraic Combinatorics, symmetric functions).
2019–2021	M.Sc. in Mathematical Sciences, Universidad of Talca, Talca, Chile.
	Título de tesis: A Waring Problem for Quadratic Forms.
	Directora: Maria Inés Icaza (Number Theory, Quadratic Forms).
2010–2017	B.Sc. in Mathematics, National University of Comahue, San Carlos de Bariloche, Argentina.
	Título de tesis: “Mixed Formulation of Eigenvalue Problems.”.
	Director: Claudio Padra. (Numerical Analysist, Finite Element Method).

Education

Skills

English Proficiency

2022	Dynamic placement test, scored C2.
2018	TOEFL, scored 109.

Computer skills

Advanced proficiency in LaTeX using Overleaf and TeXstudio.

Intermediate proficiency in Microsoft Excel.

Intermediate proficiency in Microsoft Word.

Intermediate proficiency in Moodle (LMS).

Awards

2017 Second Prize, *Concurso de monografías UMA*, for the monograph *El teorema de Lax–Milgram y su aplicación al Método de Elementos Finitos* (mathematics student monograph competition).

2016 Second Prize, *Certamen Número de Oro 2016* (university-level mathematics competition).

References

Luc Lapointe, Universidad de Talca, Chile. lapointetalca@gmail.com
Susanna Fishel, Arizona State University, United States of America. sfishel@asu.edu
Maria Inés Icaza, Universidad de Talca, Chile. icazap@utalca.cl

Updated: September 2026.